



Does public shaming deter drunk driving? A controlled interrupted time-series study

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Abstract

Objectives This study reports the one-year impact of a social media campaign aimed at deterring drunk driving in El Paso, Texas. Every Friday evening, a police department and community group uploaded booking photos of people arrested for driving while intoxicated (DWI) to social media accounts. Alcohol-related vehicle crashes were evaluated to assess the campaign's deterrent effect.

Methods Bayesian structural time-series analysis was conducted with 55 months of vehicle crash data. To isolate the campaign's effect, several control-series were incorporated, such as public transportation usage and alcohol sales, as well as crash trends in other Texas cities.

Results Overall analyses suggest the campaign was associated with a modest reduction in alcohol-related crashes. Subgroup analyses by driver age revealed strong evidence the campaign was associated with reductions among drivers ages 31 years and older. Drivers between the ages 21 to 30 showed no evidence of a reduction.

Conclusion The findings underscore both the potential and legal limits of social media campaigns as tools for crime prevention.

Keywords Deterrence · DWI · Punishment · Shame · Social media · Bayesian

Introduction

Beginning in the 1980s and continuing through the late 1990s, alcohol-related traffic deaths declined in the United States. The percentage of deaths involving a drunk driver was 48% in 1982 and fell to 30% by 1997, where it has remained for over twenty five years (National Highway Traffic Safety Administration, 2023). Under-

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standing what caused the decline is challenging given the legal and social changes that co-occurred (Williams, 2006). Between 1981 and 1986, over 700 laws aimed at drunk driving were introduced, more than 400 citizen groups were founded, and news stories about drunk driving increased fifty-fold (Hingson et al., 1988). As Wagenaar et al. (1995) concluded from an attempted meta-analysis: “[its] impossible to distinguish the separate contribution of each intervention” during this time period (p. 312). While some evaluations attempted to parse out the effects (e.g., West et al., 1989), most studies focused on legal mechanisms, such as lowering blood alcohol concentration (e.g., Zwerling & Jones, 1999), and increasing the minimum drinking age (Cook & Tauchen, 1984; McCornac, 1982; Williams et al., 1983), to name a few.

The emphasis on formal legal responses has overshadowed the role of community norms and social pressures that also contribute to crime prevention. Exploring community approaches holds particular promise: they align with broader movements encouraging communities to share responsibility for crime control with government (Garland, 2001; Clear & Karp, 2000), and they may offer new avenues for prevention strategies now that driving while intoxicated (DWI) legislation has largely stabilized nationwide. The current policy environment is therefore well-positioned to estimate the effectiveness of community-based interventions. One avenue is to examine the impact of mass media campaigns on drunk driving, an area of research where most evaluations were conducted more than two decades ago and have yet to include social media campaigns (Elder et al., 2004; Tay, 2005; Yadav & Kobayashi, 2015).

The current study adds to the DWI prevention literature by documenting the impact of a city-wide, social media campaign implemented in a single jurisdiction. Every week for one year, booking photos of people recently arrested for driving while intoxicated (DWI) were shared to social media by a police department and community group. To assess the impact the campaign had on drunk driving, a controlled interrupted time-series design is used to compare the rates of alcohol-involved vehicle crashes before and during the first year of the program. The findings show both the potential for community-based strategies, as well as the potential legal risks when community groups leverage police practices.

Prior research on DWI prevention

The study of DWI prevention began around the latter half of the 20th century, when quasi-experimental designs debuted in policy evaluation research (e.g., Berk & Rauma, 1983; Campbell & Ross, 1968; Ross et al., 1970). Several innovative methods were introduced for estimating the deterrent effect of legal sanctions where randomization was not possible (Ross & McCleary, 1983; Ross, 1984; West et al., 1989). One of the earliest methodological discussions of evaluating drunk driving interventions was provided by Ross and McCleary (1983), who argued that alcohol-related traffic crashes should serve as one of the primary outcome measures for assessing changes in drunk driving behavior. More than two decades later, this recommendation was reinforced in *Problem-Oriented Policing Guide No. 36: Drunk Driving*, which likewise identified alcohol-related vehicle crashes as one of the principal harms that DWI prevention strategies are intended to reduce (Scott et al., 2006). Consistent with

these recommendations, numerous evaluations have relied on alcohol-related crash data to assess the effectiveness of prevention initiatives, making crash-based outcomes the predominant approach in the transportation literature (Elder et al., 2002; Holder & Wagenaar, 1994).

Within the transportation literature, a body of work specifically evaluated the effects of mass media campaigns on drunk driving (Elder et al., 2004; Tay, 2005). The theories of change in this literature lack attention, although the work by Tay and Ozanne (2002) is an exception. Tay & Ozanne discussed the importance of analyzing message content in media campaigns. According to their review, prevention strategies using fear-based content often assume the content arouses fear in viewers and changes behavior, however, this is not always tested in evaluations and may show differential effects across a population (also see Fradella, 2000). For instance, in a systematic review, Tay (2005) identified five studies that estimated the deterrent effect of anti-drunk driving mass media campaigns and generally found they were all associated with reductions in alcohol-related crashes, although the mechanisms through which campaign messages influenced behavior remained unclear. While this body of research has propped media-based prevention strategies as effective prevention strategies, relatively few studies have examined the specific role of message content or communication platforms in shaping campaign effects. Moreover, despite the widespread adoption of social media as a primary channel for public health and safety messaging, evaluations of social media-based drunk driving prevention campaigns remain limited, leaving an important gap in the literature.

Another glaring challenge in the DWI prevention literature is inferring causal inference given the presence of plausible alternative explanations (Ross & McCleary, 1983). Indeed, several other factors theoretically influence the rate of traffic accidents in a jurisdiction. Numerous studies have demonstrated that enforcement activity (O'Hara et al., 2022), public transit usage (Fell et al., 2020), fuel price (Chi et al., 2011; Naqvi et al., 2020), and alcohol sales (Gruenwald & Ponicki, 1995) are correlated with alcohol-related crashes. Given that virtually all city-wide DWI prevention evaluations rely on quasi-experimental designs, the incorporation of these exogenous factors is vital to minimize bias.

The DWI friday initiative

Setting

The setting for the current study is El Paso, the sixth largest city in the state of Texas with a population of 678,815 people in an area of about 257.4 square miles (U.S. Census Bureau, 2020). El Paso is located along the southern border, adjacent to Las Cruces, New Mexico and contains four land ports of entry into Ciudad Juarez, Chihuahua, Mexico. A large military installation is located in the city with a population of 11,260 (U.S. Census Bureau, 2020). As for socioeconomic demographics, the majority of El Pasoans identify as Hispanic (81.25%) and nearly half (47.97%) are adherents of the Catholic Church (Grammich et al., 2023). In 2024, the estimated median household income was \$59,932, and the employment rate was 57.9%.

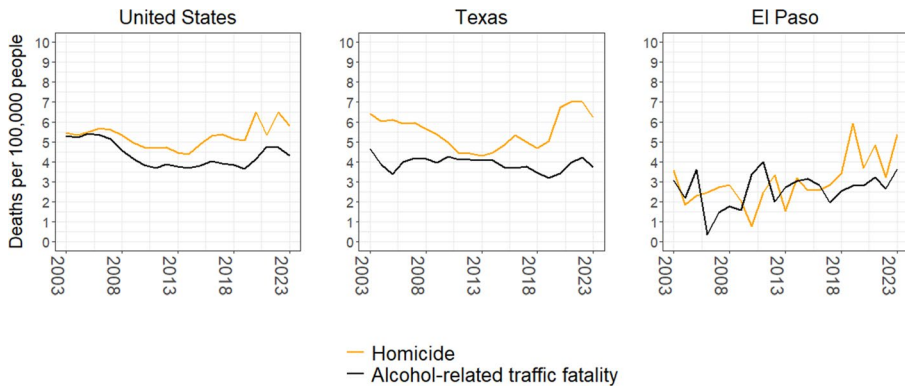


Fig. 1 Rates of homicide and alcohol-related vehicle fatalities from 2003 to 2023

With respect to alcohol-related vehicle crashes, El Paso, like other major cities in Texas, experienced a significant decline in the early 1990s. The rate of traffic fatalities was 17.14 deaths per 100,000 people in 1992 and fell to 8.31 in 2003, decreasing by over 50%.¹ Since 2003, about one-third of traffic fatalities involve alcohol as a contributing factor. The prevalence of alcohol-related traffic fatalities is nearly as prevalent as homicide in the city and is considered one of the most pressing issues facing the region (KFOX14, 2025). Figure 1 illustrates these trends over a twenty-year period in the United States, state of Texas, and city of El Paso.²

Implementation

On June 21, 2024, the El Paso Police Department (EPPD) began the “DWI Friday” initiative (Luna, 2024). Each Friday evening, the EPPD uploaded booking photos of people charged with DWI on their official Facebook and Instagram accounts. The people featured were all arrested by officers of the department during the week prior to the post. Below each booking photo, the arrestee’s name and alcohol-related charge were displayed. With the exception of Thanksgiving weekend, the EPPD consistently published DWI Friday posts since the initiative began. Over the course of 52 weeks, 1,221 photos were shared online.

The coverage and delivery of the campaign circulated through local news outlets but was systematically shared by a local community group recognized throughout the study jurisdiction for its cultural significance and ability to prompt official action (Mata, 2023; Moore & Villa, 2025). Each week immediately following the EPPD’s posts, all booking photos were obtained and reuploaded to the community group’s

¹ Fatality counts were obtained from archived Motor Vehicle Traffic Accident reports by the Texas Department of Transportation (<https://digital.library.unt.edu/explore/collections/>).

² Counts were obtained from Federal Bureau of Investigation (2023), National Highway Traffic Safety Administration (2023), and online reports from the Texas Department of Transportation (2025a). Rates were calculated using linear interpolation of decennial census data.

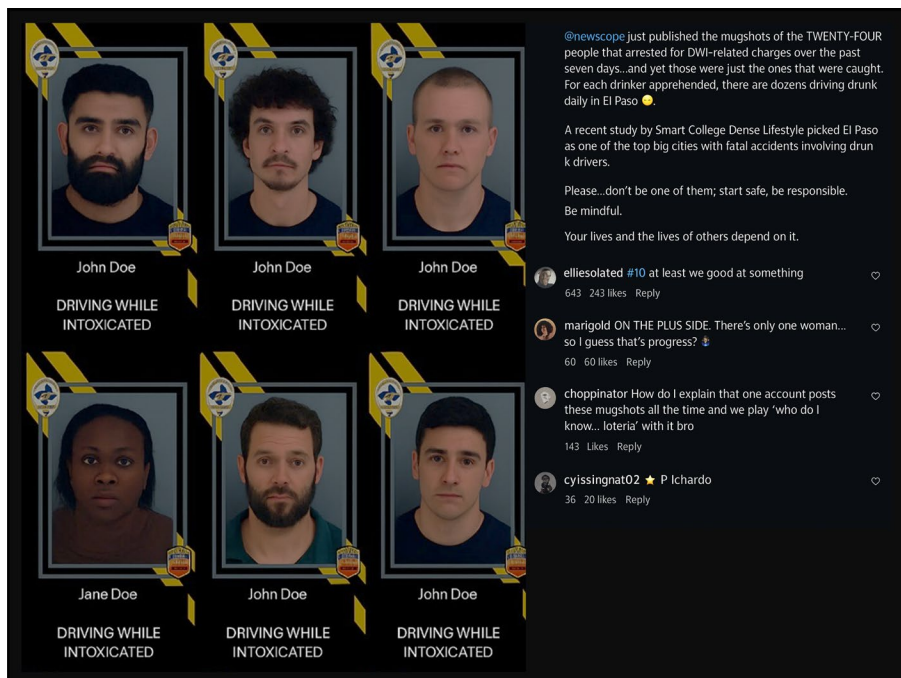


Fig. 2 Recreation of a weekly DWI Friday post

Instagram account, which could only be accessed on a computer or mobile device connected to the internet.³ Fig. 2 provides a rendition of a typical DWI Friday post created by the community group. Upon accessing the post on social media, viewers were presented with 2×3 arrangements of booking photos, along with a real-time list of comments made by Instagram users.

An important distinction between the EPPD's practice and the posts by the community social media account is the temporal access of photos and ability for viewers to read comments. The EPPD made arrest photos available for only 24 h each week and without public commenting, while the community group uploaded photos permanently and facilitated a public forum. This implementation decision significantly increased the level of exposure and determined the context in which viewers would come to interpret the campaign. During the first year, the community group's posts garnered over half a million user-likes and over 16,000 user-comments. It is important to note not all viewers clicked the "like" button. Given that no data is publicly available to assess the duration and frequency of exposure to the weekly posts, exposure to the initiative is likely underestimated but nevertheless observable using these metrics.

³The community social media account is a form of citizen journalism, where any Instagram user can send videos, photos, or messages to the account manager, who then shares the content via a permanent or temporary public post. As of July 2025, the EPPD's Facebook and Instagram accounts are followed by, respectively, over 85,000 and 39,000 users. The community account currently shows over 600,000 followers.

Additionally, the geographic residency of people who viewed the weekly posts is not publicly available, however, evidence from the comments suggests a substantial proportion of users are or were local residents in the study jurisdiction. A representative sample of 474 comments across 51 weekly posts from the community group was dichotomously coded for users who explicitly stated they recognized a person in the arrest photo.⁴ The analysis shows about 10.76% (95% CI: 8.26%, 13.90%) of all comments explicitly stated the user recognized a person in the photo (e.g., “Someone I know every week”, “I finally recognize someone!”, “That’s my neighbor”, “Damn, I know three guys on this list”).

In addition to recognizing those who were arrested, users’ comments exhibited a range of reactions to DWI Friday. Some users found the photos and comments entertaining. For example, the top comment in one week’s post contained 111 user-likes: “Literally, the best hour of the week. Here for the comments”. Other users expressed humor, sometimes generally (e.g., 235 user-likes: “yall catch everyone except my ex every week”) or towards specific people in the photos. For instance, the name of an arrested person who appeared to have strabismus appeared in twenty-five comments in one week (e.g., 921 user-likes: “I don’t think [Name] could drive sober even if he wanted too”; 438 user-likes: “[Name] didn’t see that coming”). Additionally, some comments drew on a more empathetic tone (e.g., 212 likes: “We’ve all been down in life...we should pray for their recoveries”).

Target population

Although the campaign was implemented city-wide, the delivery and timing of DWI Friday posts likely focused exposure to a subset of the population, specifically young adults between the ages 18 to 49 with access to social media. According to estimates from the Pew Research Center, about 76% of Americans between the ages of 18 to 29 use Instagram, followed by 66% of Americans between the ages of 30 to 49 (2024). The overrepresentation of young adults as Instagram users resembles the overrepresentation of young adults in El Paso who were involved in alcohol-related crashes. A descriptive analysis of crash data shows the average age of drivers in El Paso who crashed with a blood alcohol content over the legal limit was 33.38 years ($SD = 12.86$) (Texas Department of Transportation, 2025b). As for the timing of the initiative, releasing posts during Friday evenings may have impacted decision-making within hours before viewers began drinking and driving, as weekends accounted for 63.57% of all alcohol-related vehicle crashes, and over 85.77% occurred between 6:00p.m. and 5:00a.m in 2024 (Texas Department of Transportation, 2025b).

Theoretical framework

Deterrence theory is a reasonable theoretical backdrop for understanding how DWI Friday may prevent would-be offenders from drunk driving. One of the core tenets of

⁴ All initial comments (excluding replies) made between June 22, 2024, and July 5, 2025, were collected from 51 DWI Friday posts, resulting in a total of 9,484 comments. Weekly comment counts ranged from 93 to 332 ($M = 186.24$; $SD = 54.50$). To ensure fair representation across weeks, a stratified simple random sample with proportional allocation (5%) was drawn from each week, yielding the reported sample size.

crime prevention in the theory is to enhance the severity of consequences. Enhancing consequences is thought to persuade or deter people from engaging in a proscribed behavior. With regards to DWI prevention, the results from prior research suggests the deterrent effect of enhancing legal sanctions is mixed (Evans et al., 1991; McCarthy & Oesterle, 1987). An alternative pathway less examined in the deterrence literature, however, is enhancing the severity of consequences through *non*-legal mechanisms.

Criminologists Harold Grasmick & Robert Bursik (1990) were among the first scholars to reformulate deterrence theory to accommodate non-legal mechanisms. They argued that in addition to the state's threat of legal punishment, people in community serve as another potential source for deterrence. Grasmick & Bursik termed this source, "significant others", and defined them as people whose opinions are valued by a would-be offender (e.g., parents, mentors, close friends). If a would-be offender were to violate social norms, including legal ones, "they run the risk of being *embarrassed* or suffering a loss of respect" from these significant others (Grasmick et al., 1991, p. 236, emphasis added). In the parlance of deterrence theory, it is level of perceived embarrassment that enhances the consequences of committing a crime, and thus increases the likelihood of being deterred. The other deterrent mechanism described by the scholars is shame, which, unlike embarrassment, occurs without others present. Shame is a reflective punishment put on by the would-be offender themselves, defined as "a self-imposed sanction [that] occurs when [a person] violates norms they have internalized" (p. 43). According to Grasmick & Bursik, embarrassment and shame both foster the internalization of social norms and weigh into the would-be offender's decision-making. So long as the perceived consequences outweigh the perceived benefit of committing the crime, the would-be offender is generally deterred.

Grasmick & Bursik's (1990) general deterrence model closely aligns with other influential theorists during that time. In *Crime, Shame, and Reintegration*, Braithwaite (1989) emphasized the role of shaming as a preferable alternative to coercive legal sanctions, describing it as a "route to freely chosen compliance" (p. 10). Unlike Grasmick and Bursik, Braithwaite distinguished between two forms of shaming: disintegrative and reintegrative. Both express social disapproval, but disintegrative shaming focuses on the person, effectively stigmatizing them, whereas reintegrative shaming focuses on the behavior and preserves the offender's social bonds to society. Another notable contribution comes from Williams and Hawkins (1986), who underscored the critical role of information networks. They argued that community awareness of a wrongdoing is a necessary precondition for informal social control processes to be activated. Overall, these concepts related to shaming emphasize the role that perceptions and community members have on crime prevention, and they provide an appropriate backdrop for assessing media campaigns that use shame or embarrassment as a method of enhancing punishment.

When applied to DWI Friday, a perceptual deterrence model shows two pathways a person can be generally deterred from drunk driving. The first pathway operates through perceived social consequences of significant others. Upon seeing the post, the would-be offender recognizes that the public dissemination of arrest photos increases the chance that significant others will find out. Significant others—and their perceived disapproval— increase feelings of embarrassment (i.e., enhance the severity of consequences) and thus reduce the likelihood of engaging in drunk driving.

The second pathway operates through self-disapproval (i.e., shame). The mere act of viewing the post and comments reinforces the viewer's sense of law-abiding norms in the community, which eventually become their own. This internalization goes on to foster self-disapproval of the act and may even effect the viewer's perception of being caught (Piquero & Paternoster, 1998).

The weekly exposure schedule of DWI Friday is another important feature of the campaign. One possibility is the perceived fear of social consequences and internalization of norms grows stronger with every exposure, thereby moving more people to not drink and drive. The other possibility is the perceived fear of social disapproval eventually "loses its sting" and deters less people as time passes. Nagin (1998) called attention to this possibility in his discussion of deterrence effects, theorizing that the novelty or fact that so few are punished is what makes the punishment severe. According to Nagin's logic, any deterrent effect from the campaign would eventually wane over time, as viewers become more desensitized to the perceived consequences.

Current study

To assess if the DWI Friday social media campaign was indeed associated with a deterrent effect on drunk driving, the current study utilized an interrupted time-series design using alcohol-related crashes as the key outcome variable. It is important to note that the DWI Friday campaign was not implemented as a randomized experiment. Consequently, any observed reductions in alcohol-related crashes cannot, on their own, be attributed conclusively. To address this limitation, publicly available data on five correlates of alcohol-related vehicle accidents were used. These include gross alcohol sales, public transportation usage, and broader indicators of economic activity.⁵ Integrating these variables into a multi-variate analysis importantly controls for their potential influence, which yields a less biased estimate of the campaign's effect (Shadish, 2002).

Methods

Data sources

To evaluate the campaign, data from several sources were obtained. To assess the incidence of drunk driving, data were queried from the Texas Department of Transportation's Crash Records Information System to obtain all vehicle collision records that occurred in the city of El Paso between January 1, 2021 and July 31, 2025. These records represent all reportable vehicle collisions systematically collected by Texas law enforcement agen-

⁵ Enforcement activity, specifically total DWI arrests by the El Paso Police Department, was initially considered but presented an endogeneity problem for this analysis. Presumably, people involved in an alcohol-related crash are subsequently arrested for DWI. This appeared to be the case in this study, which showed monthly alcohol-related crashes and DWI arrests were positively correlated. Analyses not reported here found no substantial differences when DWI arrests were included in the analysis.

cies.⁶ All records included information such as the time and day a crash occurred, how many vehicles were involved, and the age of the drivers. The remaining sources were used to account for plausibly related measures effecting drunk driving: public transportation usage in the study jurisdiction;⁷ mean monthly retail gasoline prices reported in the state of Texas (U.S. Energy Information Administration, 2025); the monthly unemployment rate in El Paso (U.S. Bureau of Labor Statistics, 2025); and finally, the total gross alcohol sales recorded by the Texas Alcohol Beverage Commission (2025).

Outcome-series

All alcohol-related crashes served as the key outcome variable. Crashes were defined as alcohol-related if alcohol was flagged as either a contributing or possibly contributing factor for any of the involved drivers. Alcohol-related traffic fatalities was also considered, however, this outcome was too rare that the lack of variance would not be appropriate for analysis ($Range=0-3$, $M=1.08$, $SD=0.95$). Given that younger people use social media more than their older counterparts, alcohol-related crashes was disaggregated by age of the involved driver to explore the possibility that younger drivers would be more deterred by the campaign than older drivers. Thus, the following three outcomes were analyzed:

- All alcohol-related vehicle crashes.
- Alcohol-related vehicle crashes involving a driver ages 21 to 30 years old.
- Alcohol-related crashes involving a driver ages 31 and older.

The cutoff of 30 years was based on the age variable's 50th percentile in the alcohol-related crash data. This ensured there were enough observations each month to model variation and also coincidentally reflects the social media usage statistics reported by the Pew Research Center (2024). Crashes were aggregated to monthly counts and converted into per capita rates of crashes per 100,000 people using decennial census data for the city of El Paso.⁸ Fig. 3 shows the trends of all three outcome-series using nearest-neighbor moving averages.

Control-series

The evidence from prior literature suggests several exogenous factors plausibly affect the rate of alcohol-involved vehicle crashes. Below are the five control-series used to

⁶ Under Section 550.026 of the Texas Transportation Code, a vehicle collision is required to be reported to law enforcement if the operator of a vehicle involved is injured, or the damage to a vehicle renders it unable to be normally and safely driven. These reporting requirements can be expanded by municipalities (see Section 550.067 of the Texas Transportation Code), which was the case in the study jurisdiction. According to Chapter 12.16 of the City of El Paso's Code of Ordinances, drivers involved in an accident in the city of El Paso are required to report it to the police department if damage to the vehicle or property appears to meet or exceed one thousand dollars.

⁷ Data was obtained from a Freedom of Information Act request filed with the city of El Paso.

⁸ The results did not differ when using estimates of annual vehicle miles traveled for urban populations (> 5,000 residents) in El Paso County.

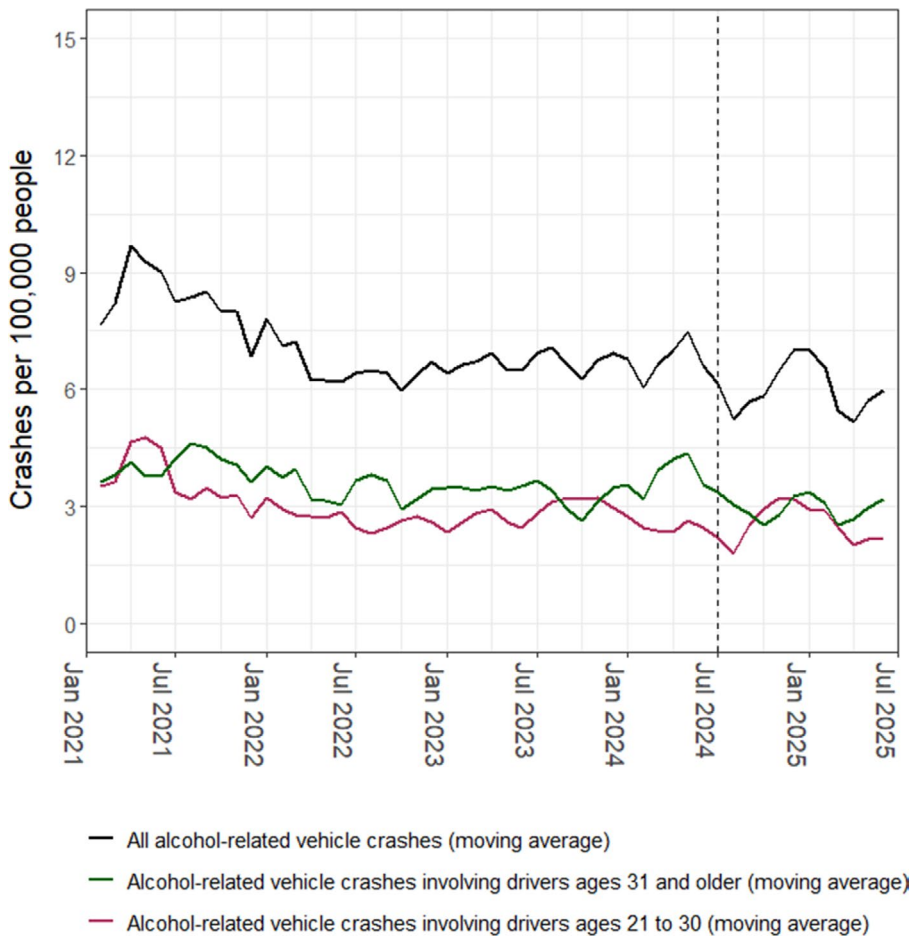


Fig. 3 Crash trends in El Paso, Texas during the observation period (pre-campaign period = 42 months; post-campaign period = 13 months)

isolate the campaign's effect. All control-series were aggregated to the month-level, as not all data were available at smaller units of analysis. Table 1 displays descriptive statistics for all variables in the analysis.

- Total riders of public transportation services.
- Mean price of conventional retail fuel in the state of Texas.
- Rate of unemployment in El Paso, Texas.
- Gross sales of beer, wine, and liquor by licensed establishments in El Paso, Texas.
- Total non-alcohol-related vehicle crashes.⁹

⁹The inclusion of non-alcohol-related crashes broadly represents exogenous factors not captured in the analysis. For instance, increased usage of rideshare services (e.g., Lyft and Uber) could have plausibly impacted both alcohol-related crashes and all other types of vehicle crashes (e.g., Arif Khan et al., 2002).

Table 1 Descriptive statistics of sample during observation period ($n=55$ months)

	Mean	(SD)	Min	Max
Outcome-series				
All alcohol-related crashes	7.11	(1.34)	5.01	11.20
Alcohol-related crashes involving drivers ages 21 to 30	3.51	(0.82)	1.77	6.19
Alcohol-related crashes involving drivers ages 31 and older	4.10	(0.79)	2.80	6.19
Control-series				
All non-alcohol-related crashes	227.55	(17.71)	181.12	262.20
Gross alcohol sales (beer, wine, liquor)	16,838,260.58	(2,181,182.40)	9,243,833.00	20,343,128.00
Rate of unemployment	4.59	(0.92)	3.70	7.60
Public transportation usage	455,398.60	(79,979.47)	235,268.00	597,997.00
Mean retail fuel price in Texas	2.97	(0.45)	2.01	4.58

Monthly total crashes were converted to rates per 100,000 people using the decennial census count for the city of El Paso

Analytic strategy

The sharp introduction of DWI Friday resembles an interrupted time-series, where statistical techniques can be used to estimate changes in the outcome before and after an intervention occurred (Shadish, 2002). To account for localized exogenous factors noted in prior literature, this study utilized a controlled interrupted time-series (CITS) design, which incorporated data from several control-series to construct a counterfactual outcome. What follows is a brief overview of Bayesian structural time series (BSTS), the analytic technique used to estimate the causal effect of the campaign. More details on the BSTS framework can be found in Brodersen et al. (2015).

BSTS is becoming a popular analytic strategy in criminal justice studies (e.g., Mourtgos et al., 2022; Adams et al., 2023; Richards et al., 2021), as it offers several strengths. First, BSTS analyses are adept for assessing how the linear combination of covariates contribute to the temporal evolution of an outcome series. As prior literature has demonstrated, there is a range of social and economic factors that plausibly influence the rate of alcohol-related crashes. As such, the inclusion of these factors as control-series is necessary for obtaining a less biased estimate of a treatment effect. Second, the BSTS approach is modular in that several complex temporal patterns can be modelled simultaneously (e.g., seasonality effects and trends). Third, they estimate effects in a fully Bayesian framework, which provides outputs that are often more intuitive and practical for decision-makers.¹⁰

¹⁰ Under a frequentist approach, a significance test below the 5% alpha level is interpreted to mean that, assuming the null hypothesis of no effect is true, there is a 5% probability of obtaining a result at least as extreme as the one observed. Bayesian analyses can be set up to yield similar inferential statements, but they offer greater flexibility by allowing decision-makers to focus directly on probabilities tied to their interest. Rather than thinking in the dichotomous terms required by the Fisher–Neyman framework of statistical significance, a police chief or city official might instead want to know the probability an intervention produced at least a 10% reduction (for guidance on applying Bayesian analyses to policy-decisions, see Mourtgos, 2025).

BSTS models are built using a two-level state-space framework consisting of an observation equation and a state equation. The observation equation describes how the observed outcome at each time point is generated from a set of underlying, unobserved components (e.g., local trend, seasonality, and regression effects) plus white noise. The state equation governs how these unobserved components evolve over time. Model parameters and latent states are estimated as probability distributions—referred to as posterior distributions—rather than single fixed values, allowing the model to capture uncertainty in the estimated trends, seasonal patterns, regression effects, and predictions. These estimates are obtained by combining prior assumptions about plausible parameter values with the observed data, resulting in a distribution that reflect their most likely values after accounting for the information contained in the data.

For this analysis, the mean of the posterior probability distribution is reported, which represents the most probable estimate of the effect. The precision of the effect is assessed by using the standard error of the posterior distribution to compute a 95% credible interval (CrI), which retains an interpretation analogous to confidence intervals in frequentist statistics. For all causal effect estimates, Bayes factors are reported to quantify the strength of evidence (see Jeffreys, 1935). Bayes factors were calculated relative to the null hypothesis of no or opposite effect, as the ratio of posterior odds favoring a positive effect, $P(\delta > 0)$, to posterior odds favoring a nonpositive effect, $P(\delta \leq 0)$. Interpretations of Bayes factors follow those proposed by Kass and Raftery (1995), where values of 1 to 3 represent anecdotal evidence; 3 to 20 positive evidence; 20 to 150 strong evidence; and > 150 as very strong evidence against the null hypothesis of no or opposite effect.

The following equations were used to model the pre-campaign period spanning January 2021 through June 2024 ($n=42$ months). Based on these estimates, a counterfactual for the post-campaign period ($n=13$) was generated. The full model is specified as follows:

$$y_t = \mu_t + \gamma_t + \mathbf{x}_t' \beta + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma^2) \quad (1)$$

$$\mu_{t+1} = \mu_t + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma_\mu^2) \quad (2)$$

where y_t is the monthly rate of alcohol-related crashes at month t ; μ_t is the local-level component, capturing slow-moving structural changes over time; γ_t is the seasonal component with a seasonal period of 4 months; and $\mathbf{x}_t'$ is a vector of control-series predictors with coefficients estimated using spike-and-slab priors. Spike-and-slab priors are a Bayesian variable-selection approach that probabilistically assigns regression coefficients to either a concentrated “spike” distribution centered near zero or a broader “slab” distribution that permits nonzero effects (George & McCulloch, 1997). Coefficients with little evidence of association with the outcome are more likely to be assigned to the spike distribution and shrunk toward zero, whereas coefficients with stronger evidence are more likely to be assigned to the slab distribution and retain nonzero effects. This approach helps identify which control series meaningfully contribute to the model while reducing overfitting and unnecessary complexity (Scott & Varian, 2015).

Preliminary diagnostics of all outcome series revealed statistically significant ($p < .05$) autocorrelations between lags 1 to 4, as well as non-stationarity in all three outcome series. To account for the autocorrelation, a seasonal component with a period of four was used to capture the recurring quarterly structure. The inclusion of a seasonal component improved model fit for all outcomes except those involving drivers ages 21 to 30. For this outcome, the autocorrelation was modeled using an autoregressive component with a lag of 2. For all three outcome series, results from Augmented Dickey-Fuller tests indicated evidence of non-stationarity in the pre-campaign period. This was confirmed by visually inspecting the series shown in Fig. 3: an initial decline can be seen between the years 2021 and 2022. This justified the inclusion of a local-level component, which can model such shifts or trends.

All models were estimated using weakly informative priors, which provided general information about the expected variability of the outcome series while allowing the observed data to primarily determine the estimated effects. Priors were specified separately for each outcome using the standard deviation of the pre-campaign crash series. Importantly, these priors did not impose constraints on the direction or magnitude of fluctuations after the intervention; rather, they served only to provide an initial estimate of the amount of variability observed before the campaign. For local-level models, the prior for the level component was informed by the pre-campaign standard deviation, while the initial state was centered around the mean of the pre-campaign period. For local linear trend models, prior information for the level and slope components was similarly derived from the observed variability in the pre-campaign period, allowing the model to estimate potential changes in the post-campaign period without restricting the trajectory of the outcome. Models were estimated using 10,000 Markov Chain Monte Carlo (MCMC) iterations, a simulation-based procedure that repeatedly evaluates possible parameter values to estimate the most plausible effects and their uncertainty. Model diagnostics, including trace plots and R-hat statistics, were examined to assess whether estimates were stable across simulations and whether the model adequately characterized uncertainty in the estimated effects.

Results are presented in two stages to facilitate interpretation of the estimated campaign effects. First, BSTS model results are presented in tabular form for all outcomes and model specifications. These tables summarize the estimated impact of the campaign by comparing the observed number of alcohol-related crashes after the campaign with the number of crashes expected in the absence of the campaign. The tables report average and cumulative absolute effects (the estimated number of crashes prevented or increased), relative effects (the proportional change from the expected counterfactual), posterior tail-area probabilities, and Bayes factors with 95% credible intervals to summarize the magnitude and uncertainty of the estimated effects. Finally, graphical BSTS results illustrate these estimates visually by displaying the observed crash series alongside the estimated counterfactual trajectory.

To strengthen causal inference, an in-time placebo test was conducted, which estimated the effect had the campaign began six months earlier than its actual implementation date (Craig et al., 2017). This falsification test evaluates whether the observed campaign effect is driven by pre-existing trends in alcohol-related crashes. Because no intervention had yet occurred during the placebo period, a statistically significant effect would undermine the causal interpretation of the primary analysis by indicating that the estimated effect may reflect factors other than the campaign.

Results

Table 2 summarizes the causal effect estimates of the BSTS analyses, and Fig. 4 illustrates these results. BSTS analyses were carried out using two model specifications: local-level only and local-level and seasonal/autoregressive. The results from the local-level and seasonal/autoregressive models are presented here to facilitate discussion. Differences between the two are emphasized.

During the post-campaign period, the observed mean crash rates were 5.98 for all alcohol-related crashes, 2.53 for those involving drivers ages 21 to 30, and 2.97 for those involving drivers ages 31 or older. In contrast, all BSTS models estimated higher counterfactual mean rates for each outcome, respectively, 7.03, 2.72, and 3.75. Expressed as a percentage, these results indicate the DWI Friday campaign is associated with crash reductions ranging from 6.1% to 19.5% across the three outcomes.

The area of the posterior distribution of causal effect estimates equal to or greater than zero is also reported in Table 2. This statistic retains a similar interpretation to *p*-values in frequentist statistics, where near-zero values or those less than a pre-specified threshold ($p < .05$) are associated with evidence against the null hypothesis of no or opposite effect.

Overall, the analysis shows positive evidence that the campaign reduced all alcohol-related crashes (BF = 8.09, mean posterior = -1.05, 95% CrI: [-2.99, 0.71]). Alcohol-related crashes involving drivers ages 21 to 30 also show positive evidence of a reduction (BF = 4.26, mean posterior = -0.18, 95% CrI: [-0.59, 0.24]). The analysis

Table 2 BSTS results using five control-series

	Average absolute effect [95% CrI]	Cumulative absolute effect [95% CrI]	Relative ef- fect[95% CrI]	Poste- rior tail-area probability of effect	Bayes fac- tor (BF)
Local-level component only					
All alcohol-related vehicle crashes	-1.02 [-2.99, 0.79]	-13.26 [-38.84, 10.32]	-13.0% [-33.3%, 15.3%]	0.12	7.33
Alcohol-related vehicle crashes involving drivers ages 21 to 30	-0.18 [-1.43, 1.11]	-2.28 [-18.59, 14.40]	0.70% [-36.2%, 77.2%]	0.37	1.70
Alcohol-related vehicle crashes involving drivers ages 31 and older	-0.72 [-1.77, 0.21]	-9.38 [-23.03, 2.69]	-17.9% [-37.4%, 7.5%]	0.06	15.67
Local-level + Seasonal/Autoregressive component					
All alcohol-related vehicle crashes	-1.05 [-2.99, 0.71]	-13.68 [-38.84, 9.20]	-13.4% [-33.3%, 13.4%]	0.11	8.09
Alcohol-related vehicle crashes involving drivers ages 21 to 30	-0.18 [-0.59, 0.24]	-2.37 [-7.65, 3.12]	-6.1% [-18.9%, 10.5%]	0.19	4.26
Alcohol-related vehicle crashes involving drivers ages 31 and older	-0.79 [-1.88, 0.11]	-10.22 [-24.49, 1.44]	-19.5% [-38.8%, 3.9%]	0.04	24.00

Interpretations of Bayes factors: 1–3 represent anecdotal evidence; 3–20 positive evidence; 20–150 strong evidence; >150 very strong evidence against the null hypothesis of no or opposite effect

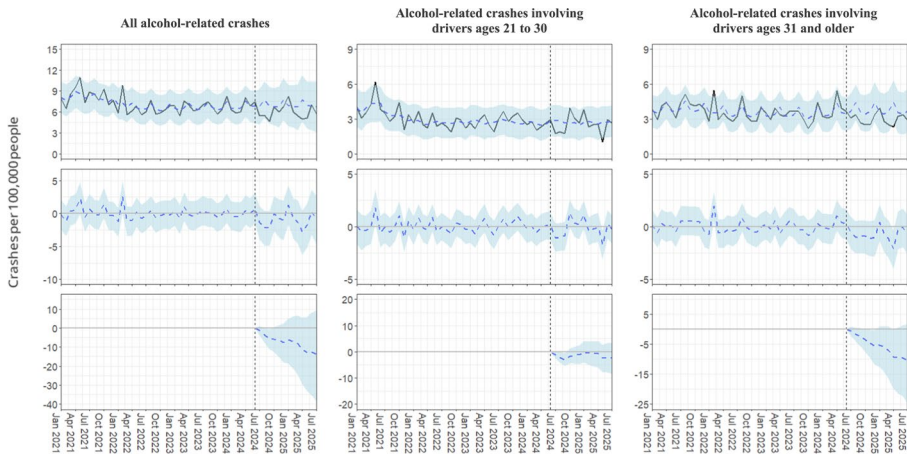


Fig. 4 Plots of results from BSTS analysis (local+seasonal/autoregressive component models). The top row displays the observed outcome series and the estimated counterfactual series in the absence of the campaign. The middle row presents the pointwise effects, representing the estimated difference between observed and counterfactual outcomes at each time point. The bottom row displays the cumulative effects, illustrating the aggregate estimated impact of the campaign over the post-campaign period. The vertical dashed line represents the month the campaign started

of alcohol-related crashes involving drivers ages 31 and older shows the strongest evidence of a decline ($BF=24.00$, mean posterior = -0.78 , 95% CrI: $[-1.88, 0.11]$). It's important to note that the posterior distribution of estimates for all three outcomes included values greater than zero. When expressed as a percentage in favor of the alternative hypothesis, the analysis shows an 89% probability the campaign caused any reduction for alcohol-related vehicle accidents and about a 96% probability the campaign caused any reduction for alcohol-related vehicle accidents involving drivers ages 31 and older. As for alcohol-related vehicle accidents involving drivers ages 21 to 30, the probability of any reduction is sensitive to model specification: the probability of any reduction is 63% for the local-level only model and increases to 79% when the autoregressive component is added.

Additional analyses

To assess the robustness of the BSTS analyses, all models were re-estimated with the campaign period beginning six months earlier (January 2025). These results are presented in Table 3. If the analyses produce similar evidence of reductions, this would suggest alcohol-related vehicle crashes were declining before the campaign began and would potentially explain the observed findings. The results here indicate a downward trend was not causing the observed decline for all alcohol-related vehicle accidents and those involving drivers ages 31 and older. The BF values for both of these outcomes decreased across both model specifications. As for alcohol-related vehicle crashes involving drivers ages 21 to 30, the probability of a reduc-

Table 3 BSTS results using control-series with false start date (six months earlier)

	Average absolute effect [95% CrI]	Cumulative absolute effect [95% CrI]	Relative ef- fect[95% CrI]	Posterior tail-area probability of effect	Bayes factor (BF)
Local-level component only					
All alcohol-related vehicle crashes	-0.74 [-3.18, 1.48]	-13.97 [-60.43, 28.20]	-7.8% [-33.8%, 31.3%]	0.24	3.17
Alcohol-related vehicle crashes involving drivers ages 21 to 30	-0.52 [-2.12, 1.00]	-9.95 [-40.31, 18.92]	-9.4% [-45.8%, 63.5%]	0.22	3.55
Alcohol-related vehicle crashes involving drivers ages 31 and older	-0.19 [-1.33, 0.96]	-3.66 [-25.30, 18.25]	-2.6% [-29.4%, 42.3%]	0.34	1.94
Local-level+ Seasonal/Au- toregressive component					
All alcohol-related vehicle crashes	-0.57 [-2.82, 1.66]	-10.74 [-53.56, 31.62]	-5.5% [-31.2%, 36.4%]	0.28	2.57
Alcohol-related vehicle crashes involving drivers ages 21 to 30	-0.21 [-0.76, 0.22]	-4.06 [-14.52, 4.20]	-7.0% [-23.2%, 9.6%]	0.16	5.25
Alcohol-related vehicle crashes involving drivers ages 31 and older	-0.07 [-1.05, 0.98]	-1.42 [-20.00, 18.6]	0.6% [-24.6%, 43.7%]	0.42	1.38

Interpretations of Bayes factors: 1–3 represent anecdotal evidence; 3–20 positive evidence; 20–150 strong evidence; >150 very strong evidence against the null hypothesis of no or opposite effect

tion increased, suggesting the observed decline among this age group was occurring before the DWI Friday campaign began.

In an effort to account for broader changes in crash patterns, a supplemental analysis was carried out using crash data from 15 Texas cities during the same observation period. Specifically, dynamic time warping, an algorithm for measuring similarity between time-series data (e.g., Eun, 2021), was used to rank potential control cities for each of the three outcomes in El Paso. Results from these analyses are presented in the [Appendix](#). Overall, the findings mirror the direction and magnitude from the previous analysis. The DWI Friday campaign was associated with a reduction for all alcohol-related vehicle crashes (BF: 6.69, mean posterior: -0.59, 95% [CrI: -1.65, 0.43]); a reduction in crashes involving drivers ages 21 to 30 (BF: 2.23, mean posterior: -0.16 [95% CrI: -0.82, 0.47]); and a reduction for crashes involving drivers ages 31 and older (BF: 32.33, mean posterior: -0.55 [95% CrI: -1.16, 0.04]). Overall, these results continue to suggest the DWI Friday campaign is associated with a deterrent effect, particularly among drivers older than 30 years old.

Discussion

Summary of findings

This study estimated a social media campaign's effect on drunk driving in a single jurisdiction. To accomplish this, alcohol-involved crash records were analyzed to

identify patterns before the campaign began. These patterns were then used to construct a predicted rate of crashes that could be compared to the observed rate. Sub-group analyses were also carried out on the age of involved drivers to understand if the campaign's effectiveness varied. To minimize the possibility that other changes in the study jurisdiction could explain the observed results, several records were obtained, specifically usage of public transportation, alcohol sales, and other economic indicators. Before reporting the results, two limitations must be acknowledged.

The first is that the campaign was not implemented under experimental conditions. This problem was handled by incorporating five potential correlates of alcohol-related crashes and conducting falsification tests to account for a pre-existing downward trend. Nevertheless, the possibility stands that an unaccounted change in the study jurisdiction may be misleading the results. The second limitation pertains to the theory of change underlying the campaign. Although social psychologists have demonstrated that public condemnation heightens the experience of humiliation and induces feelings of shame (e.g., Combs et al., 2010; Smith et al., 2002), this study did not test if these assumptions were met. Moreover, the informal qualitative review of comments suggests not all users explicitly expressed feelings of fear, shame, or embarrassment. The results and interpretations that follow must be viewed in light of these limitations.

Overall, the analysis shows DWI Friday is associated with a reduction in alcohol-involved crashes in El Paso, Texas. According to the analysis, an average of 7.13 alcohol-related crashes were prevented each month during the first year of the campaign. Additional analyses were conducted to examine whether the population most plausibly exposed to the campaign—drivers ages 21 to 30—experienced greater deterrence than drivers ages 31 years and older. Contrary to expectations, the analyses found no evidence that the campaign reduced alcohol-related crashes among drivers ages 21 to 30. Instead, the strongest evidence of deterrence was observed among drivers ages 31 years and older, for whom the campaign was associated with an average reduction of 5.36 alcohol-related crashes per month.

The contrasting findings between younger and older drivers indicate that age may shape how individuals respond to public shaming campaigns. Age has long been recognized as one of the strongest correlates of criminal and risky behavior (Hirschi & Gottfredson, 1983), and life-course perspectives suggest that increasing social bonds and adult responsibilities strengthen informal social control and increase the perceived costs of offending (Laub & Sampson, 1993). By contrast, younger adults are generally less deterrable, as they exhibit greater sensation seeking, place greater weight on immediate rewards than long-term consequences, and are less likely to perceive legal sanctions as personally consequential (Bushway et al., 2013; Tay & Ozanne, 2002). These developmental and social differences may reduce the salience of public shaming messages among younger drivers, even when they are the campaign's intended audience. Conversely, older drivers may be more responsive because the reputational, occupational, and interpersonal consequences of a DWI arrest carry greater weight, making compliance with the law more attractive. Although the present study could not directly evaluate these mechanisms, the age-graded effect observed here suggests that future research should investigate the individual-level processes through which social media campaigns influence different age groups.

Policy implications

The findings of this study carry both theoretical and practical implications for using public shaming to deter drunk driving. One theoretical implication draws from the age-graded finding. Crime prevention campaigns should not only consider “what works” but also consider “what works for whom” (Cullen, 2005). A social media campaign that is purportedly targeting a specific subgroup does not necessarily mean the target population will respond as intended. A second theoretical implication concerns the mechanism through which such campaigns are expected to influence behavior, particularly the role of embarrassment and shame in deterrence. Although these mechanisms were not explicitly tested, the informal review of comments suggests the campaign’s intended goal of eliciting embarrassment and encouraging law-abiding attitudes may be, as Maahs and Pratt (2017) observed in a similar study, “drowned out by those belonging to entertainment, emotion, and the joy that people take in the misfortune of others” (p. 56). Such variation in emotional response offers one explanation for why DWI Friday does not consistently achieve its intended effect: the content must evoke fear, not amusement, for deterrence to occur.

To this end, one could imagine modifying the content to meet this objective and hone viewers’ emotional responses. But achieving the kind of stimulus-response pattern envisioned by perceptual deterrence theory—particularly the sense of shame that amounts to “physiological discomfort”, as described by Grasmick and Bursik (1990, p. 840)—would require pushing the emotional intensity of the campaign much further. In pursuing the deterrence logic here, the ideal stimulus runs the risk of resembling a fictional archetype, such as the Ludovico treatment in *A Clockwork Orange* (Burgess, 1962) or the daily ritual of Two Minutes Hate in *1984* (Orwell, 1949). These characterizations are admittedly extreme and present obvious ethical dilemmas. Beyond attempting to elicit more fear from viewers (i.e., scare them into compliance), attempting to increase the campaign’s effectiveness in this manner raises concerns about its legality.

Arguments surrounding the constitutionality of state officials shaming offenders date back to the Supreme Court’s decision in *Paul v. Davis* (1976), in which the Court ruled that “reputation alone, *apart from some more tangible interests such as employment*,” is insufficient to trigger procedural protections (emphasis added). This principle is commonly referred to as the stigma-plus factor. In other words, social media campaigns conducted by criminal justice organizations appear to be legally permissible so long as the people depicted do not experience tangible losses. What constitutes a tangible loss, however, remains an open question. From this author’s review, one case—*Bursac v. Suozzi* (2008), decided by the New York Supreme Court—found that publicly shaming individuals accused of DWI on the internet can meet the standard of tangible harm.

The constitutional safeguard implied in *Paul v. Davis* (1976) strictly shields citizens from state officials who facilitate shaming punishments, not community groups. In the study jurisdiction, the El Paso Police Department (EPPD) is not shaming these individuals beyond what is expected from a department making public notifications of arrests. The community group, on the other hand, is the entity who shames arrested people by leveraging the EPPD’s social media practice. The implication here may

not be the legality, but the inability for the state to regulate and monitor how a community punishes its own members. The EPPD is an institution governed by law and susceptible to systematic change; a city manager can direct a police chief to terminate controversial practices. The community group, in contrast, resembles what Yale law professor James Whitman (1998) described as an “uncontrolled, general populace” (p. 1088), a group governed by evolving standards of morality that regulate how members punish. So long as the social media campaign does not result in harassment or threats of violence towards the accused, a community group is free to leverage a police department’s social media practices and embarrass moral transgressors in the name of public safety.

Thus, as the DWI Friday campaign currently stands, the person accused of DWI faces the risk of formal, regulated punishment should they be found guilty, but they are guaranteed punishment by a community group irrespective of their court outcome. How, or even if, community punishment extends beyond social media is unknown. At best, the weekly punishment ritual of DWI Friday is inconsequential—maybe even beneficial—for the offender’s reintegration, assuming they indeed committed the infraction. The other possibility is the community’s punishment extends beyond reputational harm alone, which is not afforded protection outlined in *Paul v. Davis* (1976). Future research should consider interviewing people who were posted in DWI Friday social media posts to learn if the punishment is associated with unintended consequences. Any findings that resemble tangible harms would implicate the constitutionality of the DWI Friday campaign. In another vein, future research should also assess reoffending patterns among those who were shamed, which could speak to whether social media shaming exerts a deterrent effect on recidivism outcomes.

Taken together, these ethical and potential legal implications limit the study of shaming-based interventions like DWI Friday. A promising path forward is to transform disintegrative elements of the campaign into reintegrative forms of shaming, which aim to preserve deterrent effects while minimizing ethical and practical concerns. A notable example is the Canberra RISE drinking and driving experiment, which utilizes reintegrative shaming for drunk driving offenders and has shown positive effects for reducing DWI recidivism (Tyler et al., 2007). Incorporating reintegrative strategies not only allows for safer behavioral change but also enables researchers to conduct rigorous evaluations, helping to fill critical gaps in knowledge on preventing one of the most persistent social problems in the United States.

Conclusion

This study reported the first-year impact of a social media campaign aimed at deterring drunk driving. Using alcohol-related vehicle crashes as the key outcome measure, the results show the campaign may have deterred drunk driving among drivers aged 31 and older. Drivers aged 21 to 30 showed no consistent evidence of being deterred. While social media shaming may offer some deterrent benefits, these must be balanced against ethical and legal concerns, especially when implemented by community groups without formal oversight.

Appendix

Table 4 Descriptive statistics of monthly mean crash rates in Texas cities between 01/2021 to 07/2025

Outcome	City	Mean	SD	Min	Max
All alcohol-related crashes	Amarillo	10.64	2.40	4.49	16.47
	Arlington	11.73	2.45	7.10	17.75
	Austin	10.14	1.65	7.07	14.35
	Corpus Christi	10.10	2.05	6.29	15.10
	Dallas	6.67	1.28	4.37	10.20
	Fort Worth	4.63	1.01	2.50	7.84
	Garland	6.51	2.12	0.81	11.79
	Grand Prairie	8.19	2.37	3.06	14.79
	Houston	8.99	1.45	6.16	12.67
	Irving	10.11	2.63	5.45	17.53
	Lubbock	10.56	2.35	4.28	15.94
	McAllen	11.48	3.18	4.92	18.28
	Odessa	16.35	3.53	8.74	27.09
	Plano	6.48	1.78	3.85	11.21
	San Antonio	10.67	1.68	6.83	14.15
Alcohol-related crashes involving drivers ages 21 to 30	Amarillo	4.56	1.72	0.50	9.48
	Arlington	5.32	1.50	1.78	8.62
	Austin	5.38	0.99	3.64	8.11
	Corpus Christi	4.97	1.17	1.89	8.49
	Dallas	3.24	0.73	2.07	4.75
	Fort Worth	2.05	0.56	0.87	3.81
	Garland	2.79	1.31	0.41	6.91
	Grand Prairie	3.89	1.36	1.02	6.63
	Houston	4.22	0.86	2.43	6.77
	Irving	5.02	1.56	2.73	8.57
	Lubbock	5.17	1.63	1.56	9.33
	McAllen	6.33	2.73	2.11	12.66
	Odessa	7.37	2.58	2.62	13.11
	Plano	2.97	1.06	0.70	5.60
	San Antonio	4.63	0.97	2.79	7.18
Alcohol-related crashes involving drivers ages 31 and older	Amarillo	6.04	1.60	2.00	9.48
	Arlington	7.55	1.67	4.06	11.67
	Austin	6.26	1.15	3.64	8.53
	Corpus Christi	6.22	1.68	2.83	9.75
	Dallas	4.29	0.87	2.76	7.21
	Fort Worth	2.89	0.73	1.52	5.11
	Garland	4.23	1.46	0.81	7.72
	Grand Prairie	5.36	1.88	1.02	10.20
	Houston	5.87	1.00	4.08	7.98
	Irving	6.38	1.79	1.95	10.91
	Lubbock	5.68	1.66	2.72	10.89

Table 4 (continued)

Outcome	City	Mean	SD	Min	Max
	McAllen	6.33	2.35	2.11	12.66
	Odessa	9.72	2.63	4.37	15.73
	Plano	4.42	1.45	2.10	9.46
	San Antonio	6.21	0.90	4.18	8.16

Crash rates were calculated using each city's 2020 decennial population count. Texas cities with 2,000 or greater total crashes during the observation period were used for matching. **Bolded** cities represent best matches using dynamic time warping, which was implemented using the *MarketMatching* library in R

Table 5 BSTS results using potential control cities

	Average absolute effect [95% CrI]	Cumulative absolute effect [95% CrI]	Relative effect[95% CrI]	Posterior tail-area probability of effect	Bayes Factor (BF)
Local-level component					
All alcohol-related vehicle crashes	-0.65 [-1.69, 0.40]	-8.40 [-21.93, 5.21]	-8.9% [-21.3%, 6.9%]	0.11	8.09
Alcohol-related vehicle crashes involving drivers ages 21 to 30	-0.23 [-0.86, 0.41]	-2.95 [-11.21, 5.32]	-6.0% [-22.1%, 15.5%]	0.24	3.17
Alcohol-related vehicle crashes involving drivers ages 31 or older	-0.58 [-1.18, 0.02]	-7.50 [-15.29, 0.27]	-13.3% [-24.4%, 0.6%]	0.03	32.33
Local-level component + Seasonal/Autoregressive component					
All alcohol-related vehicle crashes	-0.59 [-1.65, 0.43]	-7.61 [-21.41, 5.64]	-8.0% [-20.9%, 7.5%]	0.13	6.69
Alcohol-related vehicle crashes involving drivers ages 21 to 30	-0.16 [-0.82, 0.47]	-2.14 [-10.62, 6.12]	-4.1% [-21.1%, 18.3%]	0.31	2.23
Alcohol-related vehicle crashes involving drivers ages 31 or older	-0.55 [-1.16, 0.04]	-7.17 [-15.02, 0.48]	-12.7% [-24.1%, 1.0%]	0.03	32.33

Weak priors (0.10) for estimating the local-level component were used for all models

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Declarations

Competing interest None.

References

- Adams, I. T., Mourgos, S. M., & Nix, J. (2023). Turnover in large US policing agencies following the George Floyd protests. *Journal of Criminal Justice*, 88, 102105.
- Arif Khan, M. A., Etminani-Ghasrodashti, R., Kermanshachi, S., Rosenberger, J. M., Pan, Q., & Foss, A. (2022). Do ridesharing transportation services alleviate traffic crashes? A time series analysis. *Traffic Injury Prevention*, 23(6), 333–338.
- Berk, R. A., & Rauma, D. (1983). Capitalizing on nonrandom assignment to treatments: A regression-discontinuity evaluation of a crime-control program. *Journal of the American Statistical Association*, 78(381), 21–27.
- Braithwaite, J. (1989). *Crime, shame and reintegration*. Cambridge University Press.
- Brodersen, K. H., Gallusser, F., Koehler, J., Remy, N., & Scott, S. L. (2015). *Inferring causal impact using Bayesian structural time-series models* (pp. 247–274). *Annals of Applied Statistics*.
- Burgess, A. (1962). *A Clockwork Orange*. Heinemann.
- Bursac v. Suozzi, 22 Misc. 3d 328, 868 N.Y.S.2d 470 (Sup. Ct. Nassau County 2008)
- Bushway, S., DeAngelo, G., & Hansen, B. (2013). Deterability by age. *International Review of Law and Economics*, 36, 70–81.
- Campbell, D. T., & Ross, H. L. (1968). The Connecticut crackdown on speeding: Time-series data in quasi-experimental analysis. *Law & Society Review*, 3(1), 33–53.
- Chi, G., Zhou, X., McClure, T. E., Gilbert, P. A., Cosby, A. G., Zhang, L., & Levinson, D. (2011). Gasoline prices and their relationship to drunk-driving crashes. *Accident Analysis & Prevention*, 43(1), 194–203.
- Clear, T. R., & Karp, D. R. (2000). Toward the ideal of community justice. *National Institute of Justice Journal*, 245, 21–27.
- Combs, D. J., Campbell, G., Jackson, M., & Smith, R. H. (2010). Exploring the consequences of humiliating a moral transgressor. *Basic and Applied Social Psychology*, 32(2), 128–143.
- Cook, P. J., & Tauchen, G. (1984). The effect of minimum drinking age legislation on youthful auto fatalities, 1970–1977. *The Journal of Legal Studies*, 13(1), 169–190.
- Craig, P., Katikireddi, S. V., Leyland, A., & Popham, F. (2017). Natural experiments: an overview of methods, approaches, and contributions to public health intervention research. *Annual Review of Public Health*, 38(1), 39–56.
- Cullen, F. T. (2005). The twelve people who saved rehabilitation: How the science of criminology made a difference: The American Society of Criminology 2004 Presidential Address. *Criminology*, 43(1), 1–42.
- Elder, R., Shults, R., Sleet, D., Nichols, J., Zaza, S., & Thompson, R. (2002). Effectiveness of sobriety checkpoints for reducing alcohol-involved crashes. *Traffic Injury Prevention*, 3(4), 266–274.
- Eun, S. J. (2021). *Effects of stricter drunk-driving laws on alcohol-related road traffic death, injury, and crash rates in South Korea: A synthetic counterfactual approach using Bayesian structural time-series models* (Vol. 163, p. 106455). *Accident Analysis & Prevention*.
- Elder, R. W., Shults, R. A., Sleet, D. A., Nichols, J. L., Thompson, R. S., Rajab, W., & Task Force on Community Preventive Services. (2004). Effectiveness of mass media campaigns for reducing drinking and driving and alcohol-involved crashes: a systematic review. *American Journal of Preventive Medicine*, 27(1), 57–65.
- Evans, W. N., Neville, D., & Graham, J. D. (1991). General deterrence of drunk driving: evaluation of recent American policies. *Risk Analysis*, 11(2), 279–289.
- Federal Bureau of Investigation (2025). Crime Data Explorer: Homicide data [Data set]. U.S. Department of Justice. <https://cde.ucr.cjis.gov>
- Fell, J. C., Scolese, J., Achoki, T., Burks, C., Goldberg, A., & DeJong, W. (2020). The effectiveness of alternative transportation programs in reducing impaired driving: A literature review and synthesis. *Journal of Safety Research*, 75, 128–139.
- Fradella, H. F. (2000). Mandatory minimum sentences: Arizona's ineffective tool for the social control of driving under the influence. *Criminal Justice Policy Review*, 11(2), 113–135.
- Garland, D. (2001). *The culture of control: Crime and social order in contemporary society*. Oxford University Press.

- George, E. I., & McCulloch, R. E. (1997). Approaches for Bayesian variable selection. *Statistica Sinica*, 7(2), 339–373.
- Grammich, C., Dollhopf, E., Gautier, M., Houseal, R., Jones, D. E., Krindatch, A., Stanley, R., & Thumma, S. (2023). *2020 U.S. Religion Census: Religious congregations & membership study*. Association of Statisticians of American Religious Bodies.
- Grasmick, H. G., & Bursik, R. J. (1990). Conscience, significant others, and rational choice: Extending the deterrence model. *Law & Society Review*, 24(3), 837–861.
- Grasmick, H. G., Bursik Jr, R. J., & Kinsey, K. A. (1991). Shame and embarrassment as deterrents to non-compliance with the law: The case of an antilittering campaign. *Environment and Behavior*, 23(2), 233–251.
- Gruenewald, P. J., & Ponicki, W. R. (1995). The relationship of the retail availability of alcohol and alcohol sales to alcohol-related traffic crashes. *Accident Analysis & Prevention*, 27(2), 249–259.
- Hingson, R. W., Howland, J., & Levenson, S. (1988). Effects of legislative reform to reduce drunken driving and alcohol-related traffic fatalities. *Public Health Reports*, 103(6), 659.
- Hirschi, T., & Gottfredson, M. (1983). Age and the explanation of crime. *American Journal of Sociology*, 89(3), 552–584.
- Holder, H., & Wagenaar, A. (1994). Mandated server raining and reduced alcohol-involved traffic crashes: A time series analysis of the oregon experience. *Accident Analysis and Prevention*, 26(1), 89–97.
- Jeffreys, H. (1935). Some tests of significance, treated by the theory of probability. In *Mathematical Proceedings of the Cambridge Philosophical Society* (Vol. 31, No. 2, pp. 203–222). Cambridge University Press.
- Kass, R. E., & Raftery, A. E. (1995). Bayes factors. *Journal of the American Statistical Association*, 90(430), 773–795.
- KFOX14 (2025, May 19). El paso leaders on dwi prevention [Video]. YouTube. <https://www.youtube.com/watch?v=nWGTuWuQJtQ>
- Laub, J. H., & Sampson, R. J. (1993). Turning points in the life course: Why change matters to the study of crime. *Criminology*, 31(3), 301–325.
- Luna, M. (2024). Recent DWI arrests within El Paso. KTSM. <https://www.ksm.com/news/recent-dwi-arrests-within-el-paso/>
- Maahs, J., & Pratt, T. C. (2017). I hate these little turds! Science, entertainment, and the enduring popularity of Scared Straight programs. *Deviant Behavior*, 38(1), 47–60.
- Mata, R. (2023). El Paso's FitFam internet famous and anonymous. El Paso Inc. https://www.elpasoinc.com/news/local_news/el-paso-s-fitfam-internet-famous-and-anonymous/article_361b9c4a-6b8f-11e4-b4cf-dbf3a8b8104b.html
- McCarthy, P. S., & Oesterle, W. (1987). The deterrent effects of stiffer DUI laws: an empirical study. *Logistics and Transportation Review*, 23(4), 353.
- McCornac, D. C. (1982). The effects of government regulation on teenage motor vehicle mortality. National Bureau of Economic Research - Working Paper Series.
- Moore, R., & Villa, P. (2025). Instagram removes FitFam El Paso's page. El Paso Matters. <https://elpasomatters.org/2025/06/23/fitfam-canceled-disabled-el-paso-texas-instagram-meta/>
- Mourtgos, S. M. (2025). *Probabilities Over p-Values: A Decision Framework for Evidence-Based Policing*. CrimRxiv.
- Mourtgos, S. M., Adams, I. T., & Nix, J. (2022). Elevated police turnover following the summer of George Floyd protests: A synthetic control study. *Criminology & Public Policy*, 21(1), 9–33.
- Nagin, D. S. (1998). Criminal deterrence research at the outset of the twenty-first century. *Crime and Justice*, 23, 1–42.
- Naqvi, N. K., Quddus, M. A., & Enoch, M. P. (2020). *Do higher fuel prices help reduce road traffic accidents?* (Vol. 135, p. 105353). Accident Analysis & Prevention.
- National Highway Traffic Safety Administration (2025). Traffic Safety Facts: Annual Report tables [Data set]. U.S. Department of Transportation. <https://cdan.dot.gov/tsftables/tsfar.htm>
- National Highway Traffic Safety Administration. (2023). *Traffic Safety Facts 2023 Data: Fatal Crashes and Percentage Alcohol-Impaired Driving, by Time of Day and Crash Type (Table 31)*. U.S. Department of Transportation. <https://cdan.dot.gov/>
- O'Hara, S. E., Paschall, M. J., & Herd, D. A. (2022). Determining a dosage threshold of drink-driving enforcement operations: A systematic review. *Drug and Alcohol Review*, 41(7), 1610–1620.

- Orwell, G. (1949). 1984. London, UK: Secker & Warburg.
- Paul v. Davis, 424 U.S. 693 (1976).
- Pew Research Center (2024). Social media fact sheet [Fact sheet]. <https://www.pewresearch.org/internet/fact-sheet/social-media/>
- Piquero, A., & Paternoster, R. (1998). An application of Stafford and Warr's reconceptualization of deterrence to drinking and driving. *Journal of Research in Crime and Delinquency*, 35(1), 3–39.
- Richards, T. N., Nix, J., Mourtgos, S. M., & Adams, I. T. (2021). Comparing 911 and emergency hotline calls for domestic violence in seven cities: What happened when people started staying home due to COVID-19? *Criminology & Public Policy*, 20(3), 573–591.
- Ross, H. L. (1984). Social control through deterrence: Drinking-and-driving laws. *Annual Review of Sociology*, 10(1), 21–35.
- Ross, H. L., & McCleary, R. (1983). Methods for studying the impact of drunk driving laws. *Accident Analysis & Prevention*, 15(6), 415–428.
- Ross, H. L., Campbell, D. T., & Glass, G. V. (1970). Determining the social effects of a legal reform: The British breathalyser crackdown of 1967. *American Behavioral Scientist*, 13(4), 493–509.
- Scott, S. L., & Varian, H. R. (2015). Bayesian variable selection for nowcasting economic time series. In A. Goldfarb, S.M. Greenstein, & C. E. Tucker (Eds.), *Economic Analysis of the Digital Economy* (pp. 119–135). University of Chicago Press
- Scott, M. S., Emerson, N. J., Antonacci, L. B., & Plant, J. B. (2006). *Drunk driving*. US Department Justice, Office of Community Oriented Policing Services.
- Shadish, W. R. (2002). Revisiting field experimentation: field notes for the future. *Psychological Methods*, 7(1), 3.
- Smith, R. H., Webster, J. M., Parrott, W. G., & Eyre, H. L. (2002). The role of public exposure in moral and nonmoral shame and guilt. *Journal of Personality and Social Psychology*, 83(1), 138.
- Tay, R. (2005). Mass media campaigns reduce the incidence of drinking and driving. *Evidence-Based Healthcare and Public Health*, 9(1), 26–29.
- Tay, R., & Ozanne, L. (2002). *Who are we scaring with high fear road safety advertising campaigns?* Asia Pacific Journal of Transport. (in press).
- Texas Department of Transportation (2025a). Annual Texas motor vehicle crash statistics archive. <https://www.txdot.gov/data-maps/crash-reports-records/motor-vehicle-crash-statistics/archive.html>
- Texas Department of Transportation (2025b). Crash Records Information System (CRIS) query tool [Data set]. <https://www.txdot.gov/data-maps/crash-reports-records/crash-data-analysis-statistics.html>
- Texas Alcoholic Beverage Commission (2025). Mixed Beverage Sales Receipts [Data set]. Texas Open Data Portal. <https://data.texas.gov>
- Tyler, T. R., Sherman, L., Strang, H., Barnes, G. C., & Woods, D. (2007). Reintegrative shaming, procedural justice, and recidivism: The engagement of offenders' psychological mechanisms in the Canberra RISE drinking-and-driving experiment. *Law & Society Review*, 41(3), 553–586.
- U.S. Census Bureau (2020). 2020 decennial census: Profile of general population and housing characteristics — El Paso city, Texas [Data set]. <https://data.census.gov>
- U.S. Bureau of Labor Statistics (2025). El Paso, TX metropolitan statistical area: Economy at a glance [Dataset]. https://www.bls.gov/eag/eag.tx_elpaso_msa.htm#eag_tx_elpaso_msa.f.1
- U.S. Energy Information Administration (2025). Texas regular conventional retail gasoline prices (dollars per gallon) [Data set]. https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=p&s=emm_epmru_pte_stx_dpg&f=m
- Wagenaar, A. C., Zobeck, T. S., Williams, G. D., & Hingson, R. (1995). Methods used in studies of drink-drive control efforts: a meta-analysis of the literature from 1960 to 1991. *Accident Analysis & Prevention*, 27(3), 307–316.
- West, S. G., Hepworth, J. T., McCall, M. A., & Reich, J. W. (1989). An Evaluation of Arizona's July 1982 Drunk Driving Law: Effects on the City of Phoenix 1. *Journal of Applied Social Psychology*, 19(14), 1212–1237.
- Whitman, J. Q. (1998). What is wrong with inflicting shame sanctions? *The Yale Law Journal*, 107(4), 1055–1092.
- Williams, A. F. (2006). Alcohol-impaired driving and its consequences in the United States: The past 25 years. *Journal of Safety Research*, 37(2), 123–138.
- Williams, K. R., & Hawkins, R. (1986). Perceptual research on general deterrence: A critical review. *Law & Society Review*, 20(4), 545–572.

- Williams, A. F., Zador, P. L., Harris, S. S., & Karpf, R. S. (1983). The effect of raising the legal minimum drinking age on involvement in fatal crashes. *The Journal of Legal Studies*, 12(1), 169–179.
- Yadav, R. P., & Kobayashi, M. (2015). A systematic review: effectiveness of mass media campaigns for reducing alcohol-impaired driving and alcohol-related crashes. *Bmc Public Health*, 15(1), 857.
- Zwerling, C., & Jones, M. P. (1999). Evaluation of the effectiveness of low blood alcohol concentration laws for younger drivers. *American Journal of Preventive Medicine*, 16(1), 76–80.

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